

Survey Reporting Automation Toolkit

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Python-based workflow improvement for survey QA, variable generation, and reporting preparation

Built reusable Python workflows that automated recurring survey data prep, QA checks, variable generation, and reporting-output preparation from Forsta/Decipher exports and API inputs. The toolkit reduced reliance on fragile Excel-only logic, improved consistency across repeated deliverables, and made complex quality checks easier to rerun and reconcile.

PROBLEM AND APPROACH

Problem

Manual Excel-based logic was limiting, repetitive, and easy to break.

- Survey exports were often wide, horizontal, and messy, making multi-column validation difficult in Excel.
- Straightlining, speeders, inconsistent answers, outlier logic, and open-end keystroke checks required rules that were hard to copy reliably across files.
- Complex yearly market-sizing and valuation calculations required significant manual reconciliation when handled with workbook formulas only.

Approach

Turn recurring cleanup and QA logic into reusable Python workflows.

- Processed Forsta/Decipher exports and API inputs using Python, pandas, and numpy.
- Automated merging, reshaping, variable generation, label cleanup, and reporting-prep transformations.
- Standardized QA checks for survey quality indicators such as speeders, straightliners, inconsistent responses, reCAPTCHA v3 risk, open-end keystroke gaps, and IP/geolocation risk indicators.

WORKFLOW SHIFT

Before	Automation layer	After
Excel formulas copied across exports Manual reconciliation Higher copy/paste and version-control risk	API/export input Python validation and transformation rules Reusable scripts and workflow notes	Consistent reruns Faster QA and variable creation Cleaner reporting-ready outputs

SELECTED AUTOMATION COVERAGE AND VALUE

Coverage examples

- Survey data merging, reshaping, and normalization
- Variable generation for complex, multi-column calculations
- Straightlining, speeder, inconsistency, and outlier checks
- reCAPTCHA v3 score review, open-end keystroke reconciliation, and IP/geolocation risk flags
- Label cleanup and reporting-output preparation for downstream weighting/tabulation work

Observed value

- Reduced recurring manual preparation effort by replacing repeated Excel-only logic with reusable scripts.
- Improved consistency of QA logic across similar projects and reduced the chance of formula-copying errors.
- Reduced selected complex variable-generation work from week-level manual reconciliation to same-day script-based reruns.
- Created repeatable workflow notes and reusable logic patterns that could be adapted across similar survey reporting needs.

TOOLS AND CONFIDENTIALITY

Tools

Python · pandas · numpy · Excel · API inputs · Forsta/Decipher

Public version note

Client names, survey topics, proprietary formulas, and implementation details are anonymized. This case study focuses on reusable workflow design, survey QA logic, and reporting-prep automation.